

JEPA-FORGE Automatic feature-division selection

Validated extension | 23 datasets | September 16, 2026

Automatic context/target selection is implemented and validated across 23 public/synthetic datasets. The study evaluates 91 schema-based candidates with three seeds each: 273 additional 60-epoch GPU training runs. All 23 selections were locked before test evaluation. Only selected tasks were evaluated on the test partition (69 seed evaluations). Together with the separate earlier 138-run study, this gives 411 full GPU training runs; smoke tests are excluded.

How a feature division is chosen

The selector receives copied training/validation rows only. It compares equal context budgets and identical forecasting targets. Training labels fit classification probes, and validation labels rank candidates; JEPA training remains label-free. Candidate choice uses mean validation macro F1 or negative RMSE across all seeds, with fixed-order ties. The test partition is used only after a durable global selection lock.

Step	Decision and evidence
1. Propose	Schema templates; equal observation budgets; fixed forecasting horizon.
2. Compare	Train three GPU seeds per candidate; score validation probes.
3. Lock	Save candidate rankings, feature indices, hashes and selected checkpoints for all datasets.
4. Evaluate	Only then evaluate the selected task on held-out test rows.

The selected JEPA encoder beats raw linear on 13/23 datasets, Extra Trees on 5/23, and the random encoder on 12/23. The low-rank heuristic triggers in 198/273 candidate runs. These descriptive comparisons do not establish statistical significance or a general JEPA advantage. Unselected candidates were not scored on test, so this study does not estimate test-set improvement over every possible feature division.

Selected classification tasks

Mean +/- sample SD over three seeds. Macro F1: higher is better.

Dataset / selected context	Cols	JEPA	Raw linear	Extra Trees
Wine / last columns	6	0.899 +/- 0.015	0.891 +/- 0.000	0.936 +/- 0.014
Digits / right half	32	0.883 +/- 0.022	0.857 +/- 0.000	0.902 +/- 0.011
Breast cancer (WDBC) / worst measurements	10	0.969 +/- 0.005	0.963 +/- 0.000	0.953 +/- 0.009
Ionosphere / alternating tokens	16	0.897 +/- 0.036	0.883 +/- 0.000	0.978 +/- 0.010
Sonar / alternating tokens	30	0.730 +/- 0.014	0.712 +/- 0.000	0.776 +/- 0.014
Semeion / left half	128	0.754 +/- 0.013	0.768 +/- 0.000	0.824 +/- 0.009
Letter / last columns	8	0.665 +/- 0.036	0.598 +/- 0.000	0.823 +/- 0.004
PenDigits / alternating tokens	8	0.904 +/- 0.002	0.807 +/- 0.000	0.948 +/- 0.004
SatImage / bands 1 3	18	0.828 +/- 0.007	0.766 +/- 0.000	0.874 +/- 0.004
Human activity (HAR) / first columns	280	0.785 +/- 0.024	0.946 +/- 0.000	0.895 +/- 0.008
Dry Bean / last columns	8	0.918 +/- 0.005	0.926 +/- 0.000	0.916 +/- 0.006
ISOLET / alternating columns	308	0.764 +/- 0.012	0.943 +/- 0.000	0.932 +/- 0.012
Nonlinear multiview / first columns	32	0.881 +/- 0.005	0.840 +/- 0.000	0.907 +/- 0.010
Hierarchical classes / alternating columns	40	0.968 +/- 0.006	0.965 +/- 0.000	0.968 +/- 0.006
Sparse interactions / alternating columns	48	0.730 +/- 0.007	0.907 +/- 0.000	0.867 +/- 0.012
Manifold + nuisance / first columns	48	0.807 +/- 0.047	0.773 +/- 0.000	0.973 +/- 0.004

All baseline comparisons use the selected context. Full-information classification reference, PCA, random-encoder and persistence results are provided in the complete JSON/Markdown artifacts.

Selected forecasting tasks

Mean +/- sample SD over three seeds. RMSE in each dataset's own units: lower is better; do not compare magnitudes across datasets.

Dataset / selected context	Cols	JEPA	Raw linear	Extra Trees
Original sensors / recent past	18	0.276 +/- 0.017	0.239 +/- 0.000	0.351 +/- 0.003
Lorenz trajectories / recent past	36	0.754 +/- 0.034	2.500 +/- 0.000	0.617 +/- 0.005
Mackey-Glass delay / recent past	36	0.083 +/- 0.002	0.079 +/- 0.000	0.082 +/- 0.001
Switching VAR / recent past	48	0.178 +/- 0.001	0.180 +/- 0.000	0.177 +/- 0.000
Chirp + seasonal / recent past	36	0.226 +/- 0.021	0.191 +/- 0.000	0.220 +/- 0.001
Coupled oscillators / recent past	48	0.109 +/- 0.016	0.035 +/- 0.000	0.135 +/- 0.001
Nonlinear AR / spaced past	36	0.098 +/- 0.001	0.094 +/- 0.000	0.114 +/- 0.000

All baseline comparisons use the selected context. Full-information classification reference, PCA, random-encoder and persistence results are provided in the complete JSON/Markdown artifacts.

Validation and limits

169 local tests passed, including adversarial test-data poisoning, role/group separation, fixed-horizon budgets, checkpoint-lock checks, archive scanning, and redacted privacy findings.

Independent verification: 273 checkpoint CPU replays, 273 probe replays and 23 selected export reloads.

Maximum embedding replay error: $1.34e-05$.

These are structure-based candidate templates, not learned causal feature importance or an exhaustive search. Generic tabular blocks need domain review. Probe tuning and mask ranking share validation, so validation can be optimistic. Data are the same 23 previously studied datasets with a new predeclared split (3026); this is not independent external replication. One split, three seeds, bounded public samples, missing subject identifiers in some sources, and synthetic forecasting limit generalization.

Privacy controls and remaining boundary

The privacy audit found no credentials under the applied scans. Two historical JUnit reports contained a personal machine name; their published replacements remove that metadata. A sanitized replacement for the affected branch ancestry is prepared; main-history replacement is pending explicit approval. Gitleaks and metadata checks cover staged files, branch history, archives, and PDF text; a fake credential was correctly blocked. Local commit/push hooks and GitHub CI repeat the checks. GitHub's settings state branch protection is not enforced for this private repository's current account setup; native secret-scanning/push-protection controls were not offered in the observed settings. This cannot erase copies or platform-retained objects, or guarantee every future manual/API upload. See SECURITY_AUDIT.md for scope and final publication verification.

Reproduction and full evidence

Run `scripts/run_selection.py`, `scripts/validate_selection.py` and `scripts/build_selection_report.py`. See `configs/selection.json` and `docs/SELECTION_PROTOCOL.md`. The Markdown report lists every candidate score and the exact selected feature indices. Compressed JSON preserves complete histories, device evidence and rankings. The source repository remains private.